

LECTURE L19-20 • MODULE 3 • CO3

Outliers

Week 10 | Slides: DATA209_Module_3_Data_Quality_Engineering.pptx, pages 22–30 | Laboratory: P19-20

TOPICS COVERED

- Z-score and IQR methods
- Influence of outliers on modelling
- When an outlier is an error and when it is the finding

What an outlier is

A point far from the bulk of the data **by some stated rule** — a flag raised by a threshold you chose. Sometimes it is an error: a typo, a unit slip, a sentinel. Sometimes it is the most valuable record in the dataset. Fraud detection, fault diagnosis and rare-disease screening are fields where the outliers *are* the entire point.

An outlier is not automatically wrong and not automatically deletable. "Outlier" without a stated threshold is not a claim.

Univariate rules

METHOD	RULE	ASSUMES	FAILS WHEN		
Z-score	\	$x - \text{mean}$ \	$/ \text{SD} > 3$	roughly normal	outliers inflate the SD and mask themselves
Modified z (MAD)	$0.6745 \cdot$ \	$x - \text{median}$ \	$/ \text{MAD} > 3.5$	symmetric	heavily skewed data
IQR fence	outside $Q1 - 1.5 \cdot \text{IQR}$ or $Q3 + 1.5 \cdot \text{IQR}$	nothing	skewed data — over-flags the tail		
Percentile	beyond the 1st or 99th	nothing	always flags 2%, whether or not any exist		

Masking is the trap: with two extreme values present, the standard deviation grows enough that neither exceeds 3. Use the MAD version, which is built on the median.

On raw `pima.csv` the IQR fence flags 34 Insulin points and 29 DiabetesPedigreeFunction points — the two most skewed columns. The fence flags the tail of a skewed distribution *by construction*. That is a signal to transform, not to delete.

Multivariate detection

A 1.9 m height and a 45 kg weight are each unremarkable; together they are worth checking.

- **Mahalanobis distance** — distance from the centre accounting for covariance.

- **Isolation Forest** — random splits isolate anomalies in fewer cuts; scales well.
- **Local Outlier Factor** — compares local density to neighbours'; finds region-relative outliers.
- **Elliptic Envelope** — fits a robust Gaussian; good when the data really is elliptical.

Influence

Outliers move the mean and inflate the standard deviation. They dominate squared-error objectives, so linear regression, k-means and PCA all chase them. High-leverage points can rotate a fitted line on their own. They defeat min-max scaling by stretching the range. Medians, IQRs, rank correlations and tree-based methods are comparatively unbothered.

The decision

Investigate first — always. Check against domain limits and the source record; most decisions become obvious once you know what the value is. Then:

- **Remove** only when demonstrably an error. Record the count and the reason.
- **Treat** — winsorise, cap at a percentile, or transform. Keeps the row, limits the influence.
- **Keep** when it is a genuine extreme. Report it and use robust methods.

The settling question is empirical: **recompute your headline statistic with and without the flagged points.** If it barely moves, stop worrying about them. If it moves a lot, you have found something that needs explaining. Silent deletion is the failure mode.

KEY TERMS

Outlier	Z-score	Median absolute deviation	Masking	IQR fence	Mahalanobis distance
Isolation Forest	Local Outlier Factor	Leverage	Winsorising		

COMMON ERRORS TO AVOID

-
- Deleting flagged points without investigating them
 - Using the z-score rule on skewed or contaminated data
 - Treating IQR flags on a skewed column as errors rather than as a transform signal
 - Failing to report how many rows were removed and why

READINGS — ALL FREE TO ACCESS

-
1. scikit-learn — novelty and outlier detection
https://scikit-learn.org/stable/modules/outlier_detection.html
 2. NIST/SEMATECH — outlier detection
<https://www.itl.nist.gov/div898/handbook/eda/eda.htm>
 3. Kuhn & Johnson — Feature Engineering and Selection
<http://www.feat.engineering/>

TEST YOURSELF

1. Explain masking, and name the rule that avoids it.
2. The IQR fence flags 12% of a column. What should you suspect first?
3. Give a case where the outliers are the object of study.
4. What single test decides whether an outlier decision matters?